

Naïve Bayes and Logistic Regression

Nimisha Roy
Georgia Tech

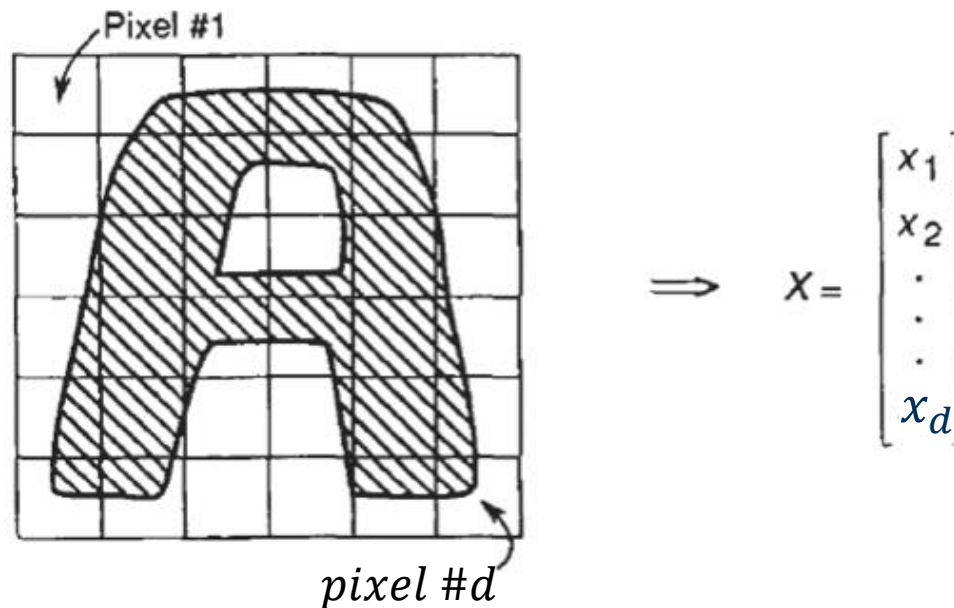
Outline

- Generative and Discriminative Classification
- The Logistic Regression Model
- Understanding the Objective Function
- Gradient Descent for Parameter Learning
- Multiclass Logistic Regression



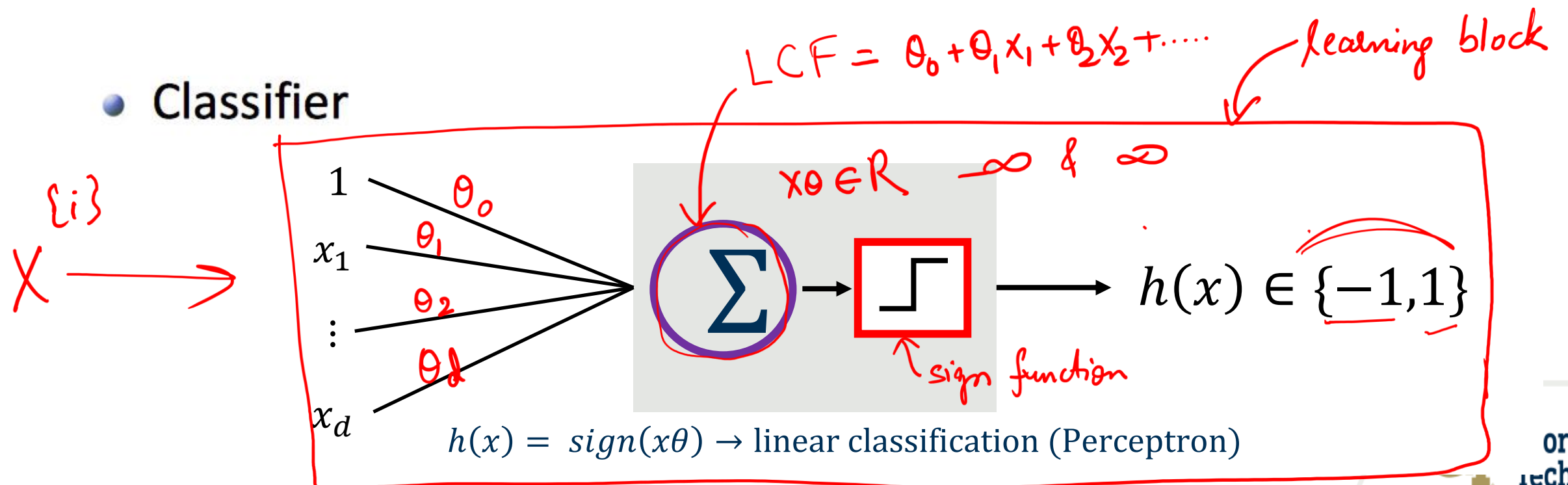
Classification

- Represent the data



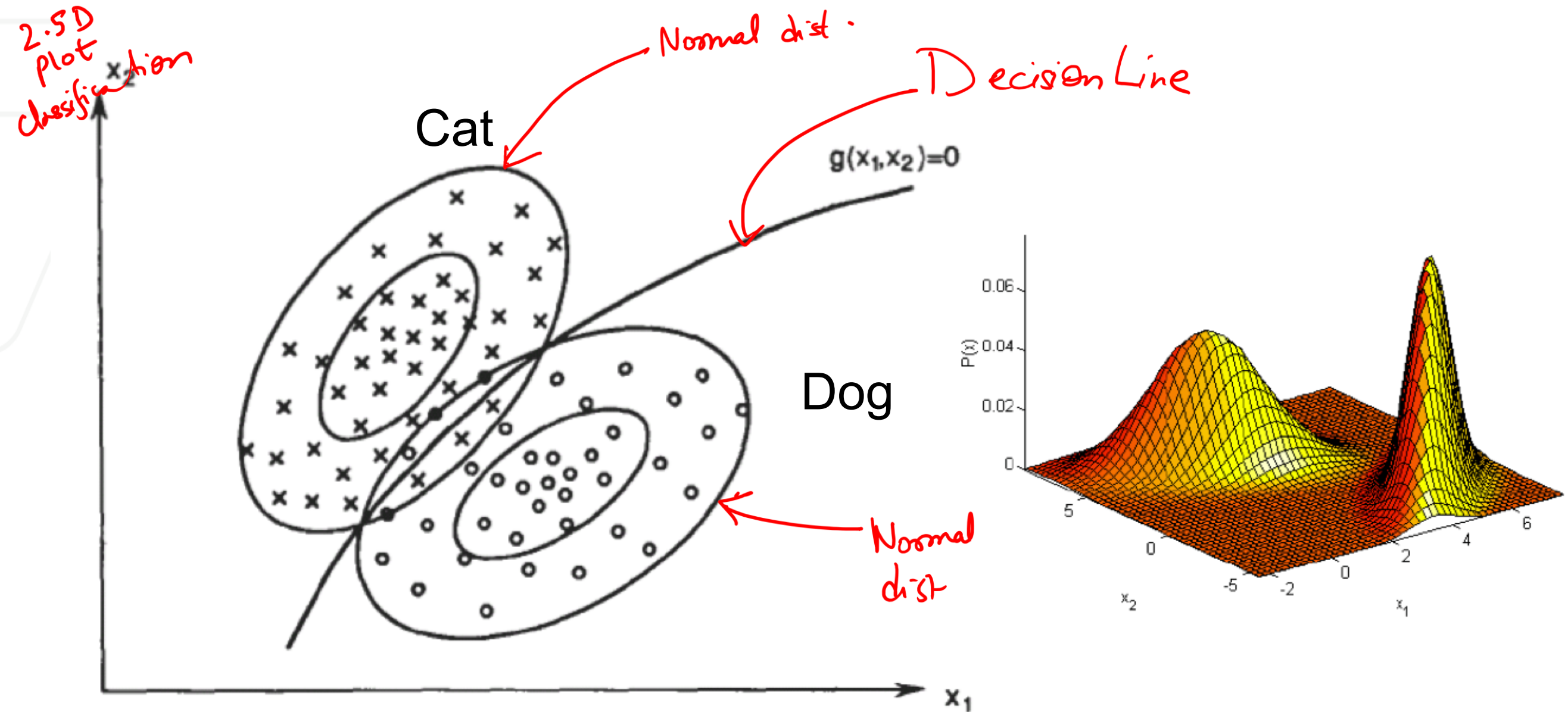
- A label is provided for each data point, eg., $y \in \{-1, +1\}$

- Classifier



Decision Making: Dividing the Feature Space

- Distributions of sample from normal (positive class) and abnormal (negative class) tissues



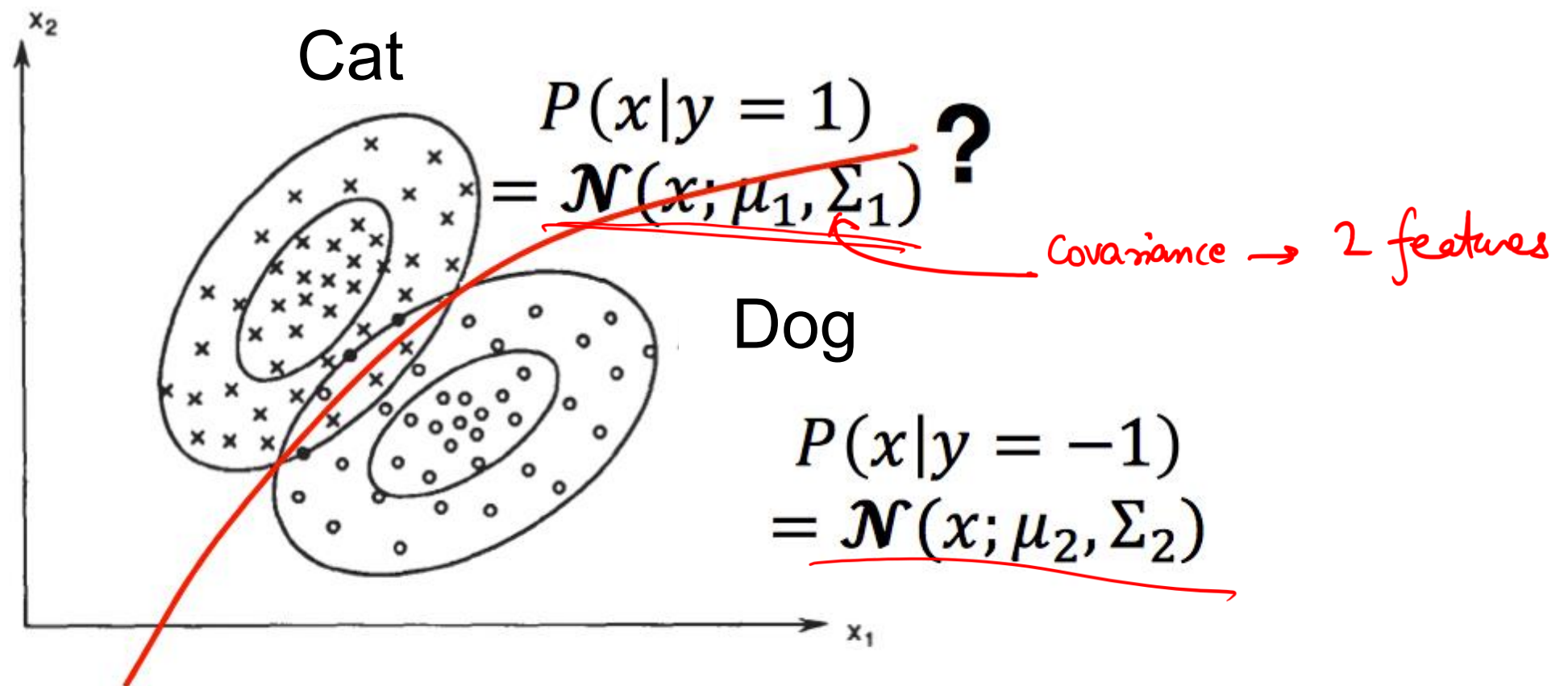
How to Determine the Decision Boundary?

$$P(y=1|x) \text{ \& } P(y=-1|x)$$

cat dog

Choose the highest one

- Given class conditional distribution: $P(x|y=1)$, $P(x|y=-1)$, and class prior: $P(y=1)$, $P(y=-1)$.



Bayes Decision Rule

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)}$$

$$P(y|x)$$

$$\left. \begin{matrix} P(y=1|x) \\ P(y=-1|x) \end{matrix} \right\} \text{choose the biggest}$$

$$P(y=1) = \frac{\text{no. of cats}}{N}$$

$$P(y=-1) = \frac{\text{no. of dogs}}{N}$$

$$P(x|y=1) = N$$

$$\begin{array}{c} \text{likelihood} \quad \text{Prior} \\ \swarrow \quad \searrow \\ P(y|x) = \frac{P(x|y)P(y)}{P(x)} = \frac{P(x,y)}{\sum_z P(x,y)} \\ \uparrow \quad \nwarrow \\ \text{posterior} \quad \text{normalization constant} \end{array}$$

Prior: $P(y)$

Likelihood (class conditional distribution : $p(x|y) =$

$$\mathcal{N}(x|\mu_y, \Sigma_y)$$

$$\text{Posterior: } P(y|x) = \frac{P(y)\mathcal{N}(x|\mu_y, \Sigma_y)}{\sum_y P(y)\mathcal{N}(x|\mu_y, \Sigma_y)}$$

$$\begin{aligned}
 P(y = \pm 1 | x^{\pm 1}) &= \frac{P(x^{\pm 1} | y = \pm 1) P(y = \pm 1)}{P(x^{\pm 1})} \\
 &= \frac{P(x_1^{\pm 1}, x_2^{\pm 1}, x_3^{\pm 1}, \dots, x_d^{\pm 1} | y = \pm 1) P(y = \pm 1)}{P(x^{\pm 1})} \\
 &= \frac{1}{2\pi^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)\right) P(y = \pm 1)
 \end{aligned}$$

Naive Bayes $\rightarrow$ assume conditional Independence of features

$$= \frac{P(x_1^{\pm 1} | y = \pm 1) \cdot P(x_2^{\pm 1} | y = \pm 1) \cdot \dots \cdot P(x_d^{\pm 1} | y = \pm 1) P(y = \pm 1)}{P(x^{\pm 1})}$$

We move from computing inverse of covariance matrix which is computationally expensive to a bunch of products of univariate gaussian distributions

Bayes Decision Rule

- Learning: prior: $p(y)$, class conditional distribution : $p(x|y)$

- The poster probability of a test point

$$q_i(x) := P(y = i|x) = \frac{P(x|y)P(y)}{P(x)}$$

- Bayes decision rule:

- If $q_i(x) > q_j(x)$, then $y = i$, otherwise $y = j$

- Alternatively:

- If ratio $l(x) = \frac{P(x|y=i)}{P(x|y=j)} > \frac{P(y=j)}{P(y=i)}$, then $y = i$, otherwise $y = j$

- Or look at the log-likelihood ratio $h(x) = -\ln \frac{q_i(x)}{q_j(x)}$

“Naïve” conditional independence assumption

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)} = \frac{P(x, y)}{P(x)}$$

Joint probability model:

$$\begin{aligned} P(x, y_{label=1}) &= P(x_1, \dots, x_d, y_{label=1}) = P(x_1 | x_2, \dots, x_d, y_{label=1}) P(x_2, \dots, x_d, y_{label=1}) \\ &= P(x_1 | x_2, \dots, x_d, y_{label=1}) P(x_2 | x_3, \dots, x_d, y_{label=1}) P(x_3, \dots, x_d, y_{label=1}) \\ &= \dots \\ &= P(x_1 | x_2, \dots, x_d, y_{label=1}) P(x_2 | x_3, \dots, x_d, y_{label=1}) \dots \\ &\quad P(x_{d-1} | x_d, y_{label=1}) P(x_d | y_{label=1}) P(y_{label=1}) \end{aligned}$$

Naïve Bayes assumption: let's rewrite it as:

$$P(x, y_{label=1}) = P(x_1 | y_{label=1}) P(x_2 | y_{label=1}) \dots P(x_d | y_{label=1}) P(y_{label=1}) =$$
$$P(y_{label=1}) \prod_{i=1}^d P(x_i | y_{label=1}) \longrightarrow \begin{array}{l} \text{Gaussian naïve Bayes} \\ \text{A typical assumption} \end{array}$$

Example

Generative Model: Naive Bayes

$$P(x, y) = \underbrace{P(x|y)}_{\substack{\text{likelihood} \\ \checkmark}} \underbrace{P(y)}_{\substack{\text{prior} \\ \checkmark}}$$

$P(y|x)$

- Use Bayes decision rule for classification

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)}$$

- But assume $p(x|y = 1)$ is fully factorized : Dimensions are conditionally independent.

$$p(x|y = 1) = \prod_{i=1}^d p(x_i|y = 1)$$

- Or the variables corresponding to each dimension of the data are independent given the label

What do People do in Practice?

- Generative models

- Model prior and likelihood explicitly
- “Generative” means able to generate synthetic data points
- Examples: Naive Bayes, Hidden Markov Models

$$P(y)$$
$$P(x|y)$$

Assume conditional independence of features
Assume gaussian

$$P(y|x)$$

- Discriminative models

- Directly estimate the posterior probabilities
- No need to model underlying prior and likelihood distributions
- Examples: Logistic Regression, SVM, Neural Networks

$$P(y|x)$$

DIRECTLY
COMPUTE

Discriminative Models

- Directly estimate decision boundary $h(x) = -\ln \frac{q_i(x)}{q_j(x)}$ or posterior distribution $p(y|x)$
 - Logistic regression, Neural networks
 - Do not estimate $p(x|y)$ and $p(y)$
- Why discriminative classifier?
 - Avoid difficult density estimation problem 
 - Empirically achieve better classification results

Generative model

Outline

- Generative and Discriminative Classification
- The Logistic Regression Model ←
- Understanding the Objective Function ←
- Gradient Descent for Parameter Learning ←
- Multiclass Logistic Regression

Gaussian Naïve Bayes

Binomial Naïve Bayes
Gaussian Naïve Bayes

$$P(\overset{\text{cat}}{y} = 1|x) = \frac{P(x|y = 1)P(y = 1)}{P(x)} = \frac{\overset{\text{prior}}{P(y = 1)} \prod_{i=1}^d \overset{\text{each feature conditional independent}}{P(x_i|y = 1)} \overset{\text{likelihood}}{}}{P(x)}$$

$$\prod_{i=1}^d p(x_i|y = 1, \mu_{1i}, \sigma_{1i})$$
$$= \prod_{i=1}^d \frac{1}{\sqrt{2\pi}\sigma_{1i}} \exp\left(-\frac{1}{2\sigma_{1i}^2} (x_{1i} - \mu_{1i})^2\right)$$

$$\text{Prior: } p(y = 1) = \underline{\pi_1}$$

Posterior: $p(y = 1 | x, \mu, \sigma, \pi)$

$$= \frac{\pi_1 \prod_{i=1}^d \frac{1}{\sqrt{2\pi}\sigma_{1i}} \exp\left(-\frac{1}{2\sigma_{1i}^2} (x_i - \mu_{1i})^2\right)}{\sum_{\substack{k=1 \\ \text{labels}}}^2 \pi_k \prod_{i=1}^d \frac{1}{\sqrt{2\pi}\sigma_{ki}} \exp\left(-\frac{1}{2\sigma_{ki}^2} (x_i - \mu_{ki})^2\right)}$$

get $\exp(\ln(u))$ of numerator and denominator $\rightarrow u$

$$= \frac{\exp\left(-\sum_{i=1}^d \left(\frac{1}{2\sigma_{1i}^2} (x_i - \mu_{1i})^2 + \log \sigma_{1i} + \underline{C}\right) + \underline{\log \pi_1}\right)}{\sum_{k=1}^2 \exp\left(-\sum_{i=1}^d \left(\frac{1}{2\sigma_{ki}^2} (x_i - \mu_{ki})^2 + \log \sigma_{ki} + C\right) + \log \pi_k\right)}$$

$$\frac{a}{a+b}$$

$$\frac{P(x|y)P(y)}{P(x|y=1)P(y=1) + P(x|y=-1)P(y=-1)}$$

$$= \frac{\exp \left(- \sum_{i=1}^d \left(\frac{1}{2\sigma_i^2} (x_i - \mu_{1i})^2 + \log \sigma_i + C \right) + \log \pi_1 \right)}{\sum_{k=1}^2 \exp \left(- \sum_{i=1}^d \left(\frac{1}{2\sigma_i^2} (x_i - \mu_{ki})^2 + \log \sigma_i + C \right) + \log \pi_k \right)}$$

$$\frac{a/a}{(a+b)/a}$$

$$= \frac{1}{1 + b/a}$$

$$= \frac{1}{1 + \exp \left(- \underbrace{\sum_{i=1}^d \left(x_i \frac{1}{\sigma_i} (\mu_{1i} - \mu_{2i}) \right)}_{\sum_i \theta_i x_i} + \underbrace{\frac{1}{\sigma_i^2} (\mu_{1i}^2 - \mu_{2i}^2)}_{\theta_0} \right) + \log \frac{\pi_2}{\pi_1} \right)}$$

$$= \frac{1}{1 + \exp(-\theta_0 + \sum \theta_i x_i)}$$

$$\sum_i \frac{1}{\sigma_i^2} (\mu_{1i}^2 - \mu_{2i}^2) + \log \frac{\pi_2}{\pi_1}$$

$$P(y = 1|x) = \frac{1}{1 + \exp \left(- \sum_{i=1}^d \left(x_i \frac{1}{\sigma_i} (\mu_{1i} - \mu_{2i}) + \frac{1}{\sigma_i^2} (\mu_{1i}^2 - \mu_{2i}^2) \right) + \log \frac{\pi_2}{\pi_1} \right)}$$

Number of parameters:

$2d + 1 \rightarrow d$ mean, d variance, and 1 for prior

$$P(y = 1|x) = \frac{1}{1 + \exp[-(\sum_{i=1}^d (\theta_i x_i) + \theta_0)]} = \frac{1}{1 + \exp(-s)}$$

$\frac{1}{1 + \exp(-x\theta)}$

Number of parameters = $d + 1 \rightarrow \theta_0, \theta_1, \theta_2, \dots, \theta_d$

Why not directly learning $P(y = 1|x)$ or θ parameters?

Gaussian Naïve Bayes is a subset of logistic regression

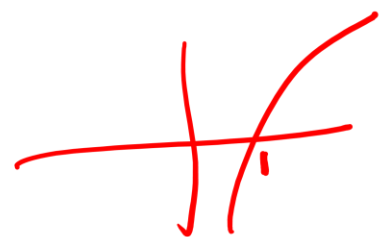
Why $\frac{1}{1+\exp(-x\theta)}$ is a probability?

$$0 \leq \text{prob} < 1$$

$$0 < \text{odds} < \infty$$

$\frac{P(y = 1|x)}{1-P(y = 1|x)}$ is called Odds

$$\frac{P(y=1|x)}{P(y=-1|x)}$$

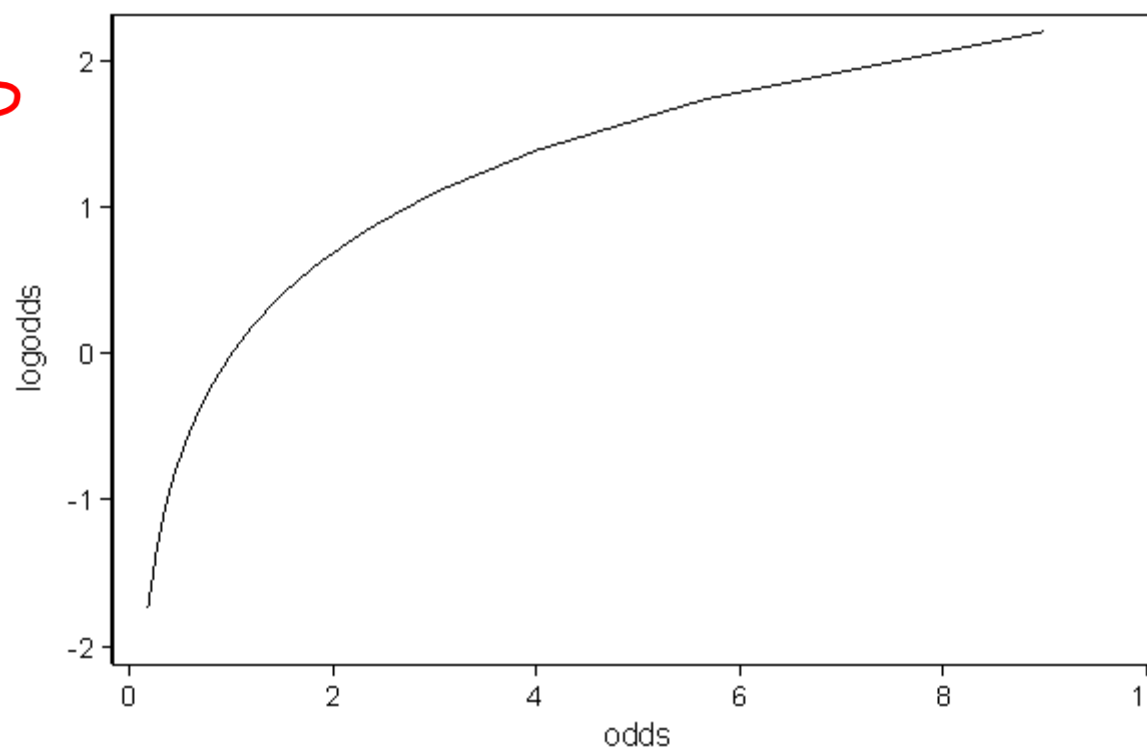
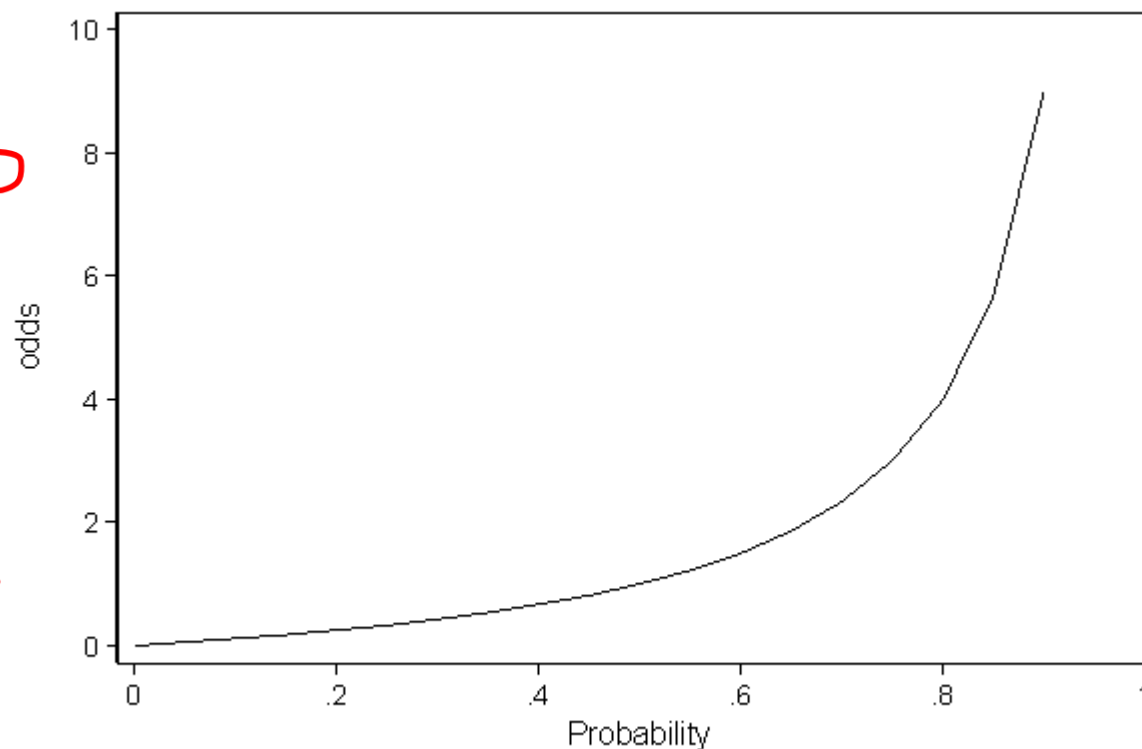


$$-\infty < \log(\text{odds}) < \infty$$

$\log(\text{odds})$ vs odds

What could be $x\theta$ domain?

$$-\infty < x\theta < \infty$$



What is logit function?

$$\text{logit}(p) = \log(\text{odds}) = \log\left(\frac{p}{1-p}\right)$$

Let's Assume

$$\log\left(\frac{p}{1-p}\right) = \theta_0 + \theta_1 x_1 + \dots + \theta_d x_d = \sum_{i=0}^d x_i \theta_i = x\theta$$

$$\exp\left(\log\left(\frac{p}{1-p}\right)\right) = \exp(x\theta)$$

Sigmoid function -
maps any real-valued
 $x\theta$ to a probability
between 0 and 1.

$$p = \frac{e^{x\theta}}{1 + e^{x\theta}} = \frac{1}{1 + e^{-x\theta}}$$



Logistic function for posterior probability

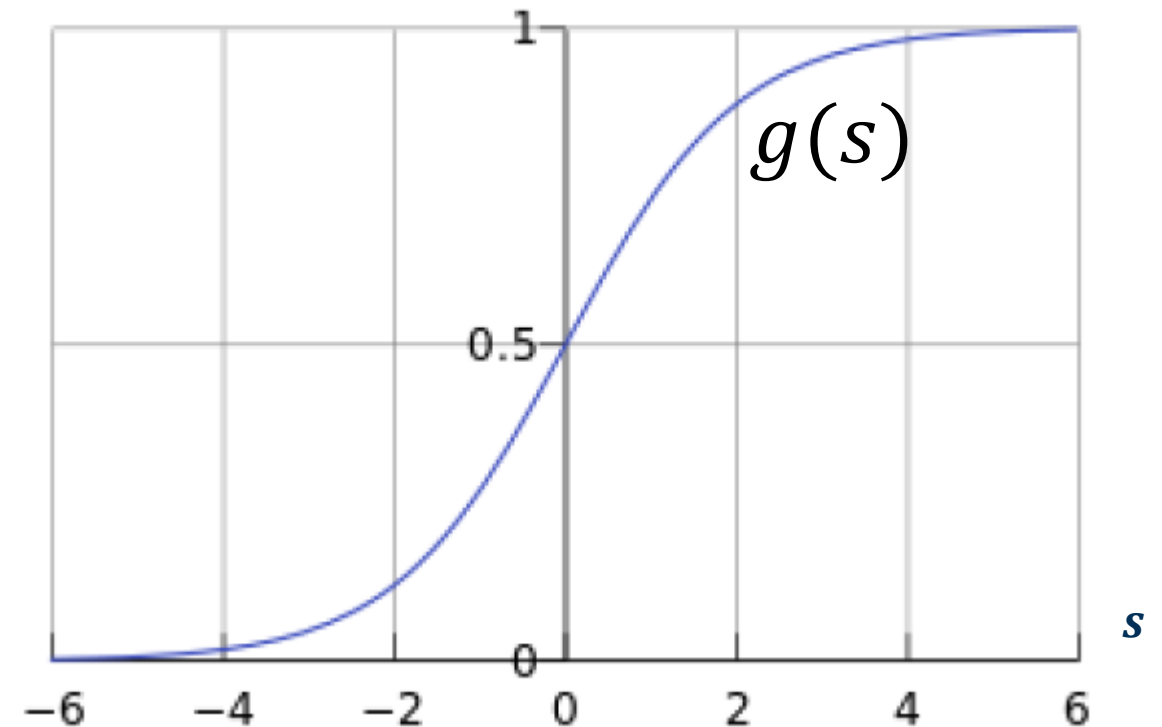
Many equations can give us this shape

Let's use the following function:

$$s = x\theta$$

$$g(s) = P(y = 1|x) = \frac{e^s}{1 + e^s} = \frac{1}{1 + e^{-s}}$$

This formula is called sigmoid function



It is easier to use this function for optimization

Is 0.5 threshold cut-off a good choice?

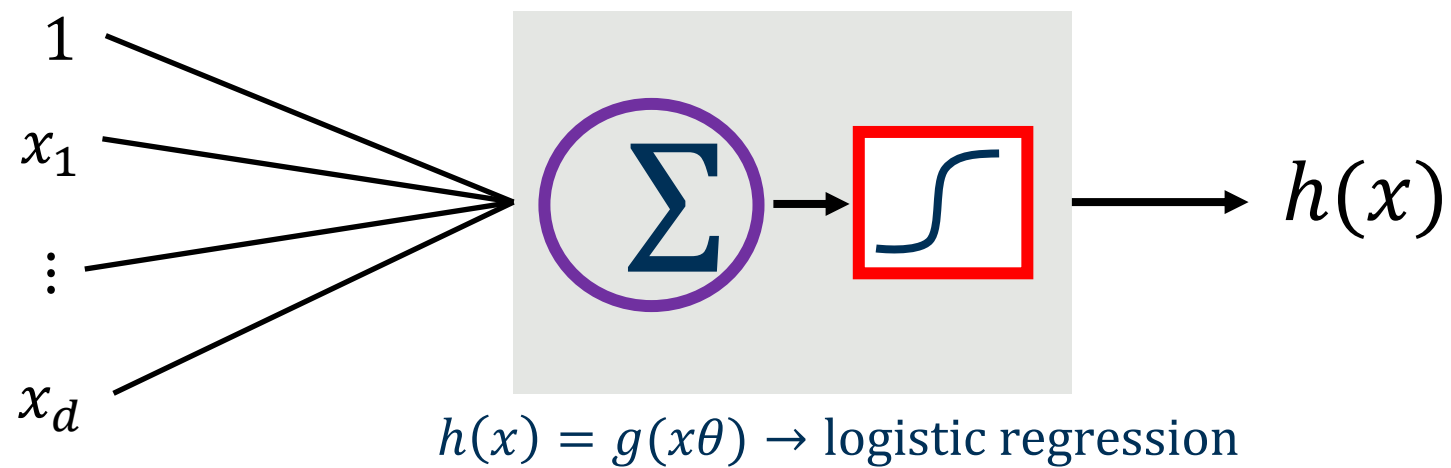
[Learn about ROC and AUC \(False positive rate and True positive rate\)](#)

([Interactive](#))

$$g(s) = \frac{e^s}{1 + e^s} = \frac{1}{1 + e^{-s}}$$

$$s = \sum_{i=0}^d x_i \theta_i = \theta_0 + \theta_1 x_1 + \cdots + \theta_d x_d$$

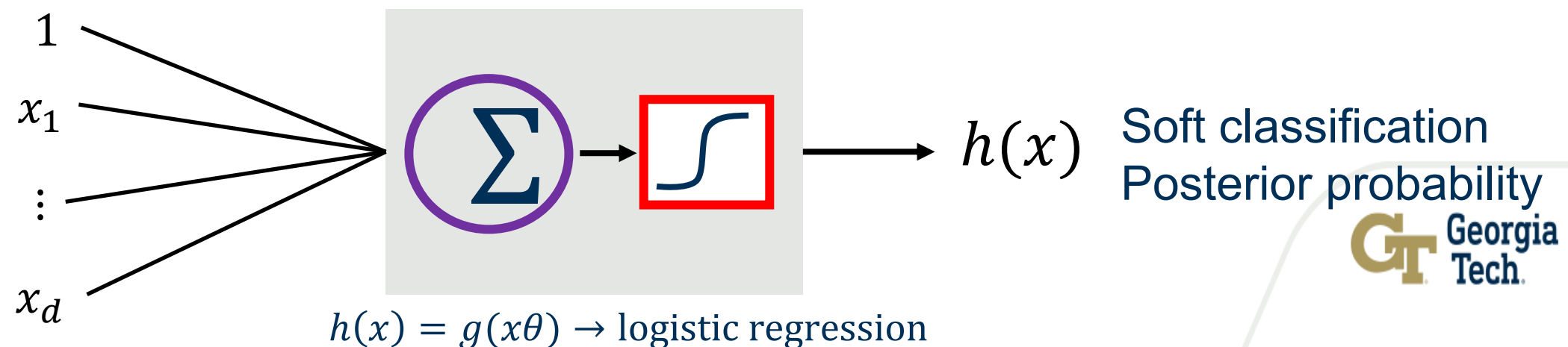
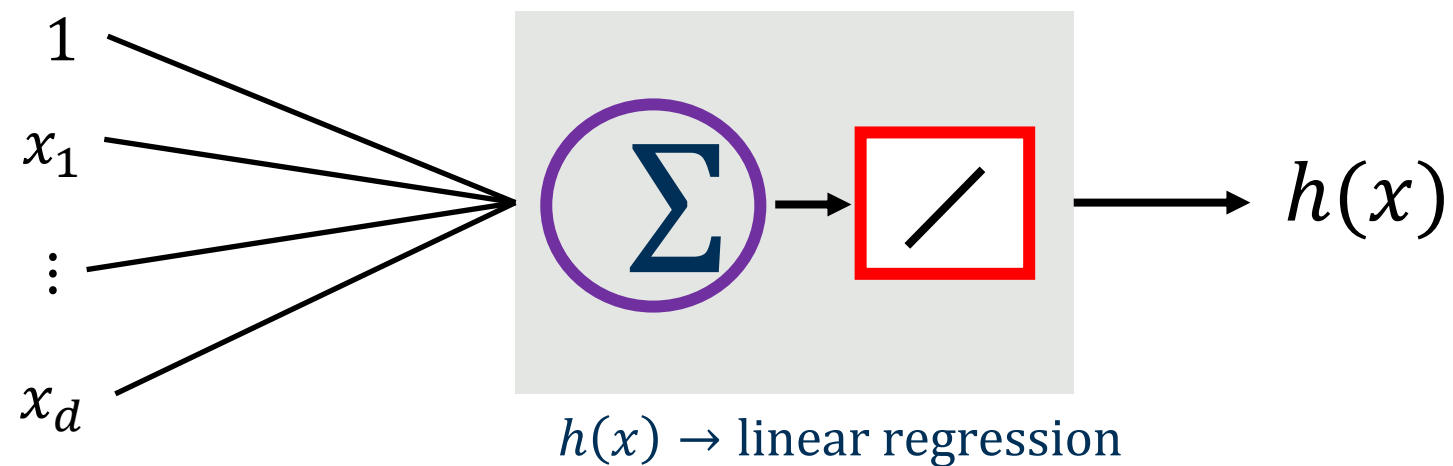
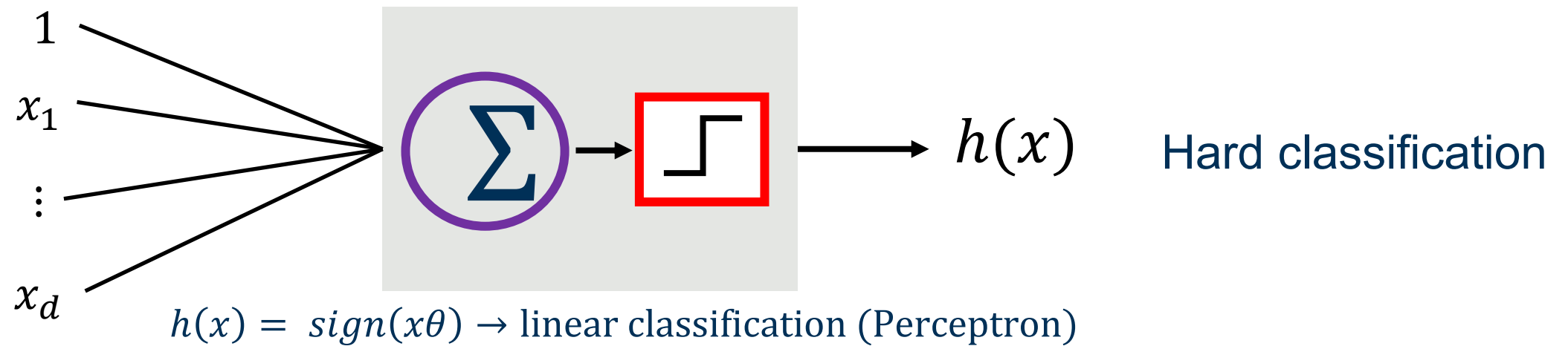
Sigmoid Function



Soft classification
Posterior probability

$$s = \sum_{i=0}^d x_i \theta_i = \theta_0 + \theta_1 x_1 + \cdots + \theta_d x_d$$

Three linear models



$g(s)$ is interpreted as probability

Example: Prediction of heart attacks

Input x : cholesterol level, age, weight, finger size, etc.

$g(s)$: probability of heart attack within a certain time

$s = x\theta$ Let's call this risk score

We can't have a hard prediction here

$$h_{\theta}(x) = p(y|x) = \begin{cases} g(s), & y = 1 \\ 1 - g(s), & y = 0 \end{cases}$$

Using posterior probability directly

Logistic regression model

$$p(y|x) = \begin{cases} \frac{1}{1 + \exp(-x\theta)} & y = 1 \\ 1 - \frac{1}{1 + \exp(-x\theta)} = \frac{\exp(-x\theta)}{1 + \exp(-x\theta)} & y = 0 \end{cases}$$

We need to find θ parameters, let's set up log-likelihood for n datapoints

$$\begin{aligned} l(\theta) &:= \log \prod_{i=1}^n p(y^{\{i\}} | x^{\{i\}}, \theta) \\ &= \sum_i \theta^T (x^{\{i\}})^T (y^{\{i\}} - 1) - \log(1 + \exp(-x^{\{i\}} \theta)) \end{aligned}$$

This form is concave, negative of this form is convex

The gradient of $l(\theta)$

$$\begin{aligned} l(\theta) &= \log \prod_{i=1}^n p(y^{\{i\}} | x^{\{i\}}, \theta) \\ &= \sum_i \theta^T (x^{\{i\}})^T (y^{\{i\}} - 1) - \log(1 + \exp(-x^{\{i\}} \theta)) \end{aligned}$$

- Gradient

$$\frac{\partial l(\theta)}{\partial \theta} = \sum_i (x^{\{i\}})^T (y^{\{i\}} - 1) + (x^{\{i\}})^T \frac{\exp(-x^{\{i\}} \theta)}{1 + \exp(-x^{\{i\}} \theta)}$$

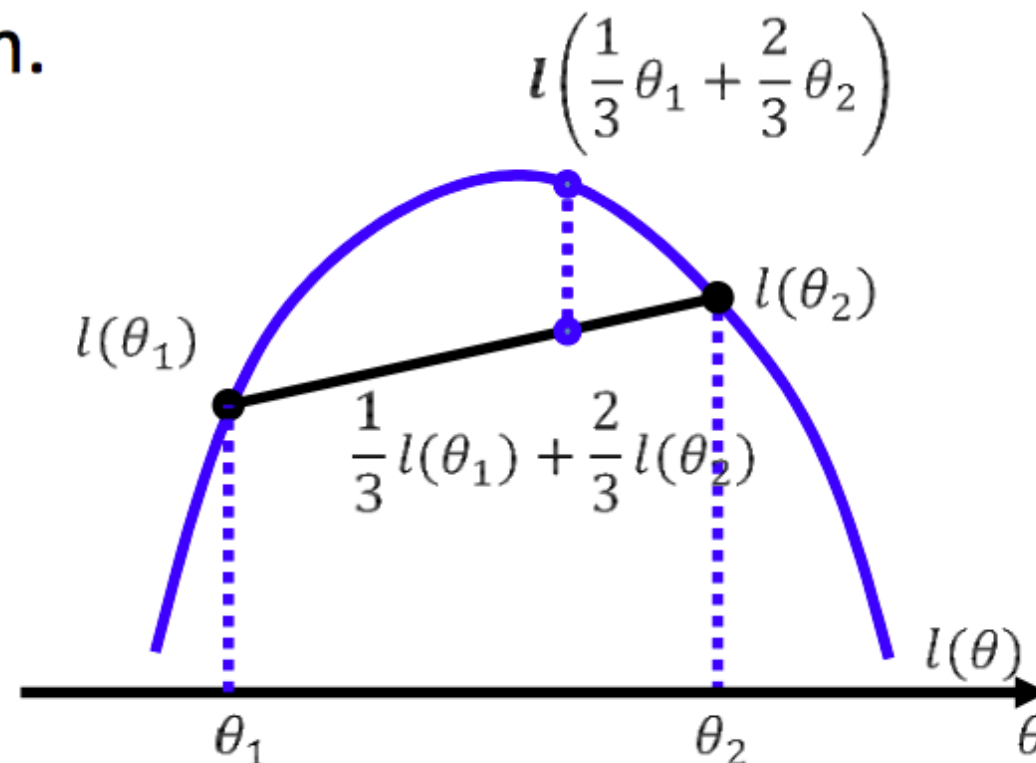
- Setting it to 0 does not lead to closed form solution

The Objective Function

- Find θ , such that the conditional likelihood of the labels is maximized

$$\max_{\theta} l(\theta) := \log \prod_{i=1}^{\bar{n}} p(y^{\{i\}} | x^{\{i\}}, \theta)$$

- Good news: $l(\theta)$ is concave function of θ , and there is a single global optimum.



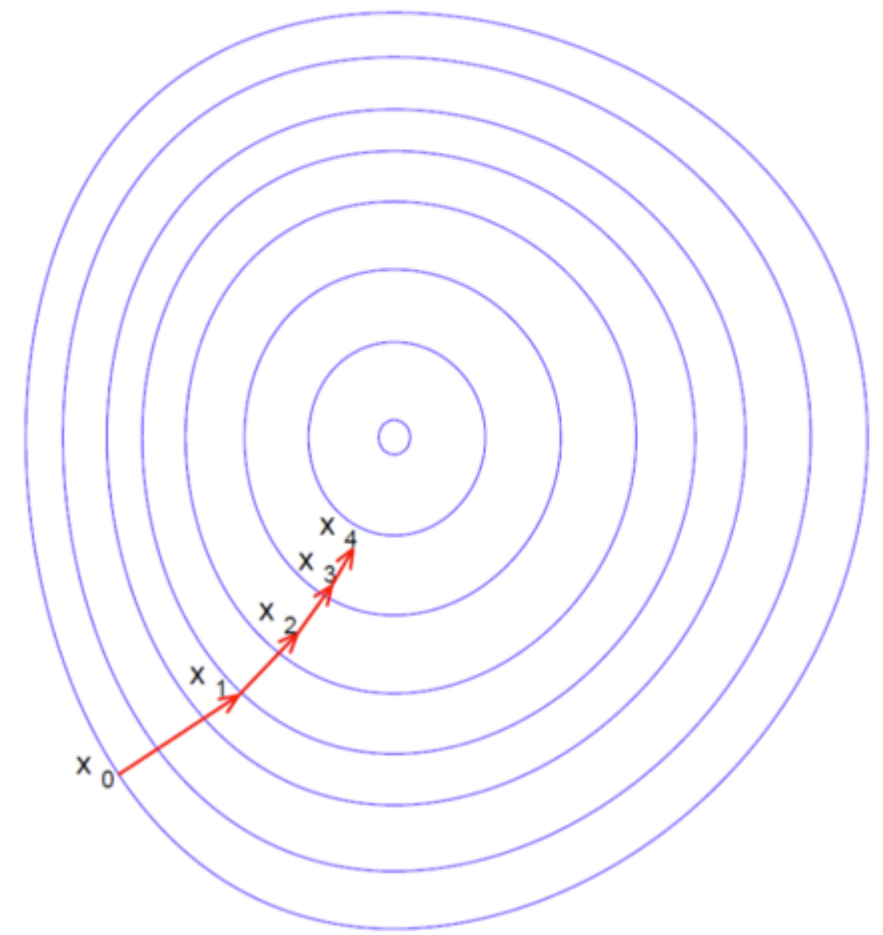
- Bad new: no closed form solution (resort to numerical method)

Gradient Descent

- One way to solve an *unconstrained* optimization problem is gradient descent
- Given an initial guess, we *iteratively* refine the guess by taking the direction of the negative gradient
- Think about going down a hill by taking the steepest direction at each step
- Update rule

$$x_{k+1} = x_k - \gamma_k \nabla f(x_k)$$

γ_k is called the step size or learning rate



Gradient Ascent(concave)/Descent(convex) algorithm

- Initialize parameter θ^0

- Do

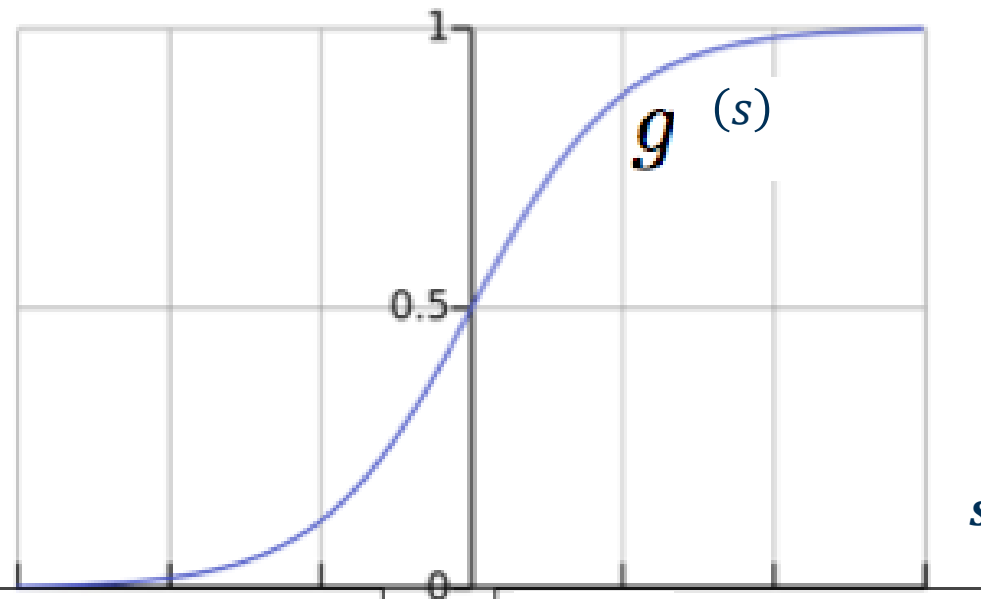
$$\theta^{t+1} \leftarrow \theta^t + \eta \sum_i (x^{\{i\}})^T (y^{\{i\}} - 1) + (x^{\{i\}})^T \frac{\exp(-x^{\{i\}}\theta)}{1 + \exp(-x^{\{i\}}\theta)}$$

- While the $||\theta^{t+1} - \theta^t|| > \epsilon$

Logistic Regression

$$g(s) = \frac{e^s}{1 + e^s} = \frac{1}{1 + e^{-s}}$$

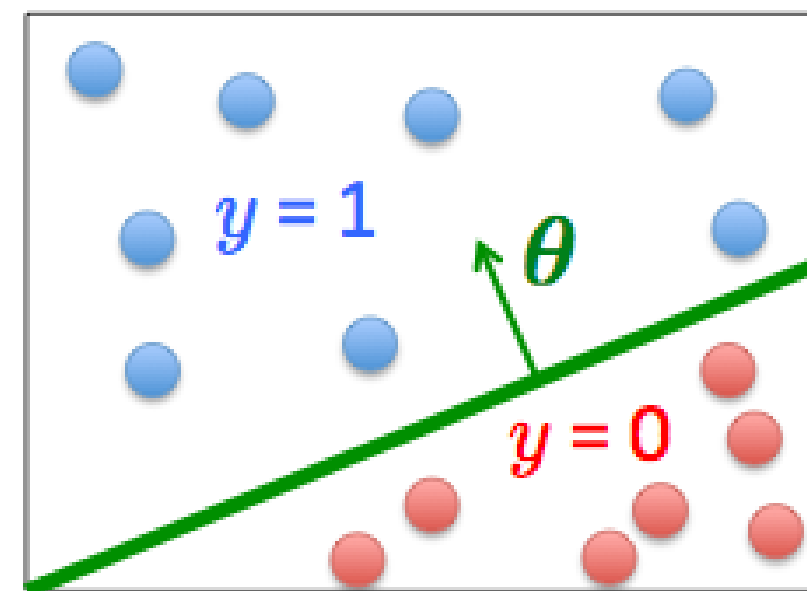
$s = x\theta$



$x\theta$ should be large negative values for negative instances

$x\theta$ should be large positive values for positive instances

- Assume a threshold and...
 - Predict $y = 1$ if $g(s) \geq 0.5$
 - Predict $y = 0$ if $g(s) < 0.5$

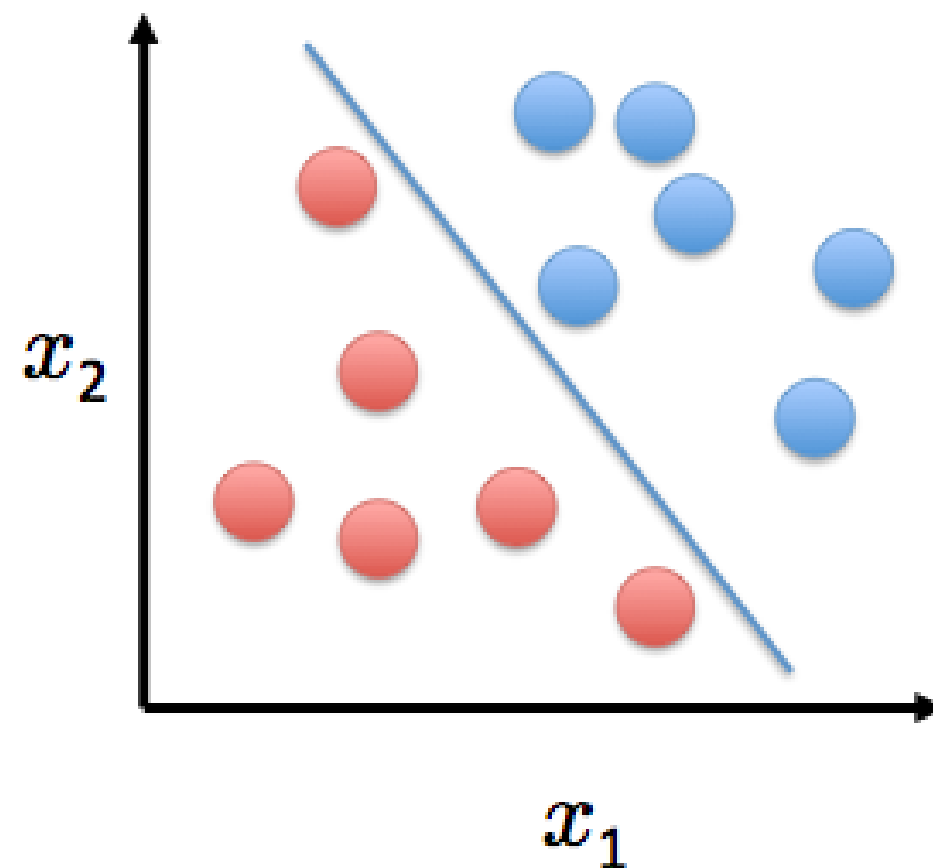


Outline

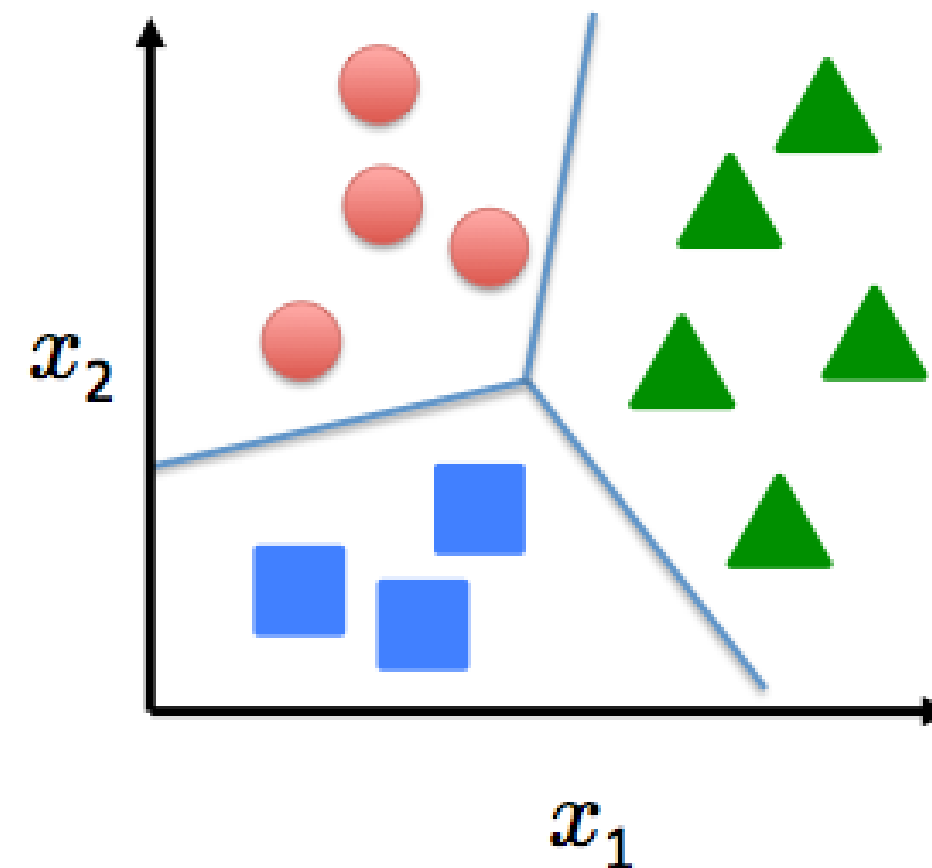
- Generative and Discriminative Classification
- The Logistic Regression Model
- Understanding the Objective Function
- Gradient Descent for Parameter Learning
- Multiclass Logistic Regression ←

Multiclass Logistic Regression

Binary classification:



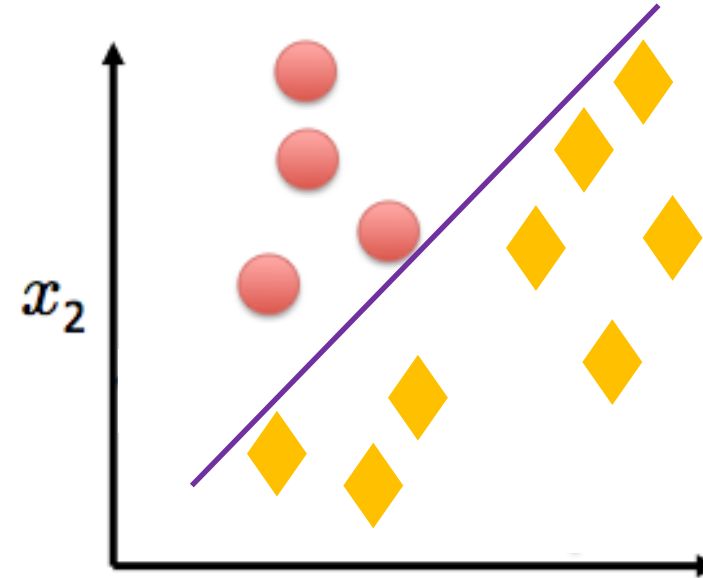
Multi-class classification:



Disease diagnosis: healthy / cold / flu / pneumonia

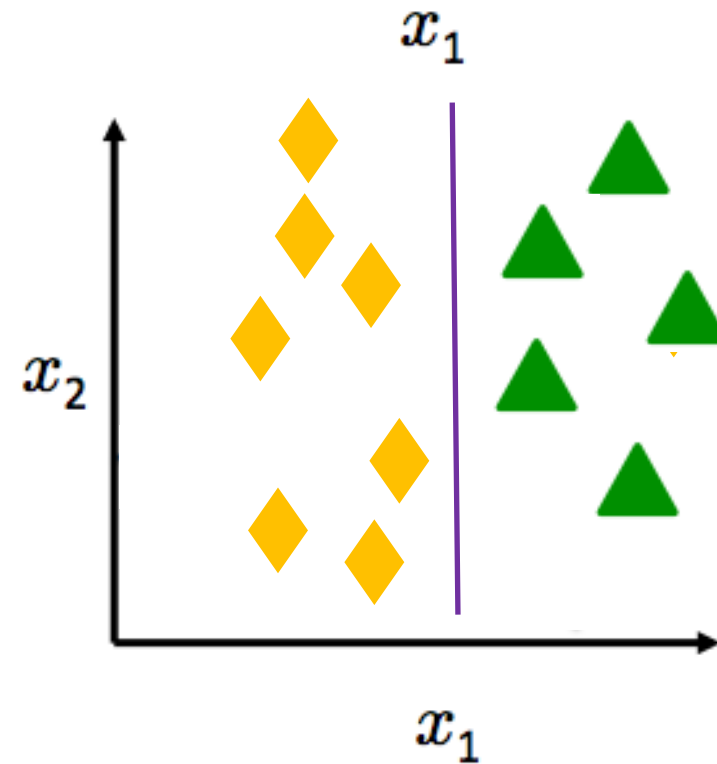
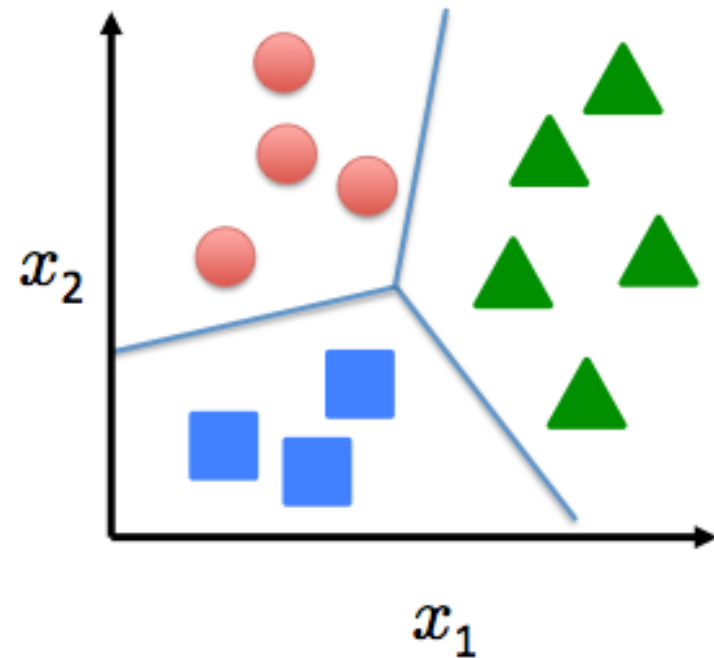
Object classification: desk / chair / monitor / bookcase

One-vs-all (one-vs-rest)

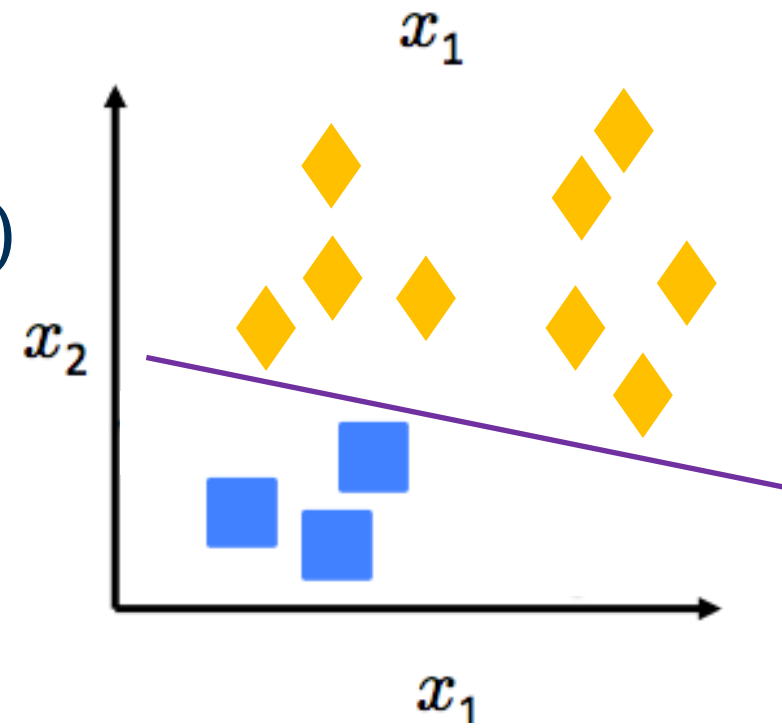


$$h_{\theta}^1(x)$$

Multi-class classification:



$$h_{\theta}^2(x)$$



$$h_{\theta}^3(x)$$

$$h_{\theta}^{(m)}(x) = p(y = 1|x, \theta) \quad (m = 1, 2, 3)$$

One-vs-all (one-vs-rest)

Train a logistic regression $h_{\theta}^{(m)}(x)$ for each class m

To predict the label of a new input x , pick class m that maximizes:

$$\max_i h_{\theta}^{(m)}(x)$$

Using Softmax

$$L(\theta) = - \sum_{i=1}^N y_a^{\{i\}} * \log(y_p^{\{i\}})$$

$$y_a = [cat, dog, fish] = [1, 0, 0]$$

$\Rightarrow$ there are M classes ($M = 3$ in this example)

$$y_p \text{ for class } \mathbf{m} = \text{softmax}(x\theta) = \frac{\exp(x\theta)_{\mathbf{m}}}{\sum_{j=0}^M \exp(x\theta)_j}$$

$$y_p = [0.6, 0.3, 0.1]$$

$$SGD \Rightarrow \theta^{t+1} \leftarrow \theta^t - \alpha \nabla L(\theta)$$

$$\theta^{t+1} \leftarrow \theta^t - \alpha x^T (y_p - y_a)$$

Take-Home Messages

- Generative and Discriminative Classification
- The Logistic Regression Model
- Understanding the Objective Function
- Gradient Descent for Parameter Learning
- Multiclass Logistic Regression

Quick Knowledge Check